

Introduction to Non-Linear Regression

There are many problems in engineering and the sciences where the response variable and the predictor variables are related through a known nonlinear function.

This leads to a nonlinear regression model.

Linear Regression Models

In previous chapters we have concentrated on the **linear regression model** defined as

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \varepsilon \quad (16)$$

The above model is called linear regression models because they are linear in the unknown parameters the $\beta_j, j = 1, 2, \dots, k$

Non-Linear Regression

Any model that is not linear in the unknown parameters is a nonlinear regression model.

The general nonlinear regression model is given as:

$$y = f(x, \theta) + \varepsilon \quad (17)$$

where θ is a $p \times 1$ vector of unknown parameters and ε is an uncorrelated random error with $E(\varepsilon) = 0$ and $Var(\varepsilon) = \sigma^2$

We also assume that the errors are normally distributed since:

$$\begin{aligned} E(y) &= E[f(x, \theta) + \varepsilon] \\ &= f(x, \theta) \end{aligned} \quad (18)$$

Non-Linear Least Squares

Suppose that we have a sample of n observations on the response and the regressors say $y_i, x_{i1}, x_{i2}, \dots, x_{ik}, i = 1, 2, \dots, n$.

The method of least squares involves minimizing the least-squares function

$$S(\beta) = \sum_{i=1}^n \left[y_i - \left(\beta_0 + \sum_{j=1}^k \beta_j x_{ij} \right) \right]^2 \quad (19)$$

Now consider the case for the non-linear case. The model is

$$y_i = f(x_i, \theta) + \varepsilon, i = 1, 2, \dots, n$$

where $x'_i = [1, x_{i1}, x_{i2}, \dots, x_{ik}]$

The least squares function is:

$$S(\theta) = \sum_{i=1}^n \left[y_i - f(x_i, \theta) \right]^2 \quad (20)$$

To find the least-squares estimates we must differentiate the above equation w.r.t θ .

This will provide a list of p normal equations for the non-linear regression situation.

The normal equations are:

$$\sum_{i=1}^n \left[y_i - f(x_i, \theta) \right] \left[\frac{\partial f(x_i, \theta)}{\partial \theta_j} \right] = 0, j = 1, 2, \dots, p$$

In a nonlinear regression model the derivatives in the large square brackets will be functions of the unknown parameters. Furthermore, the expectation function is also a nonlinear function, so the normal equations can be very difficult to solve.

Example Consider the following non-linear regression model

$$y = \theta_1 e^{\theta_2 x} + \varepsilon \quad (21)$$

The least-squares normal equation for this model are:

$$\sum_{i=1}^n \left[y_i - \hat{\theta}_1 e^{\hat{\theta}_2 x_i} \right] e^{\hat{\theta}_2 x_i} = 0 \quad (22)$$

$$\sum_{i=1}^n \left[y_i - \hat{\theta}_1 e^{\hat{\theta}_2 x_i} \right] \hat{\theta}_1 x_i e^{\hat{\theta}_2 x_i} = 0 \quad (23)$$

After simplification the normal equations are:

$$\sum_{i=1}^n y_i e^{\hat{\theta}_2 x_i} - \hat{\theta}_1 \sum_{i=1}^n e^{2\hat{\theta}_2 x_i} = 0 \quad (24)$$

$$\sum_{i=1}^n y_i x_i e^{\hat{\theta}_2 x_i} - \hat{\theta}_1 \sum_{i=1}^n x_i e^{2\hat{\theta}_2 x_i} = 0$$

These equations are not linear in $\hat{\theta}_1$ and $\hat{\theta}_2$ and no simple closed-form solution exists. In general, iterative methods must be used to find the values of $\hat{\theta}_1$ and $\hat{\theta}_2$.

Transformation to a Linear Model

It is sometimes useful to consider transformation that induces linearity in the model expectation function.

For example consider the model

$$\begin{aligned} y &= f(x, \theta) + \varepsilon \\ &= \theta_1 e^{\theta_2 x} + \varepsilon \end{aligned} \quad (25)$$

Since $E(y) = f(x, \theta) = \theta_1 e^{\theta_2 x}$ we can linearize the expectation function by taking logarithms

$$\ln E[y] = \ln \theta_1 + \theta_2 x$$

Therefore it is tempting to consider rewriting the model as:

$$\begin{aligned} \ln y &= \ln \theta_1 + \theta_2 x + \varepsilon \\ &= \beta_0 + \beta_1 x + \varepsilon \end{aligned} \quad (26)$$

A nonlinear model that can be transformed to an equivalent linear form is said to be intrinsically linear.

Parameter Estimation in a NonLinear System

Linearization

A method widely used in computer algorithms for nonlinear regression is linearization of the nonlinear function followed by the Gauss–Newton iteration method of parameter estimation.

Linearization is accomplished by a Taylor Series expansion of $f(x, \theta)$ about the point $\theta'_0 = [\theta_{10}, \theta_{20}, \dots, \theta_{p0}]$ with only the linear terms retained.

This yields:

$$f(x_i, \theta) = f(x_i, \theta_0) + \sum_{j=1}^n \left[\frac{\partial f(x_i, \theta)}{\partial \theta_j} \right] (\theta_j - \theta_{j0})$$

If we set:

$$f_i^0 = f(x_i, \theta_0)$$

$$\beta_j^0 = \theta_j - \theta_{j0}$$

$$Z_{ij}^0 = \left[\frac{\partial f(x_j, \theta_0)}{\partial \theta_j} \right]_{\theta=\theta_0}$$

We note that the nonlinear regression model can be written as

$$y_i - f_i^0 = \sum_{j=1}^p \beta_j^0 Z_{ij}^0 + \varepsilon_i, i = 1, 2, \dots, n$$

That is we now have a linear regression model. We call θ_0 the starting values for the parameters.

We may write the above equation as

$$\mathbf{y}_0 = \mathbf{Z}_0 \beta_0 + \epsilon$$

so the estimate of β_0 is

$$\hat{\beta}_0 = (\mathbf{Z}_0' \mathbf{Z}_0)^{-1} \mathbf{Z}_0' \mathbf{y}_0 = (\mathbf{Z}_0' \mathbf{Z}_0)^{-1} \mathbf{Z}_0' (\mathbf{y} - \mathbf{f}_0)$$

Now since $\beta_0 = \theta - \theta_0$, we could define

$$\hat{\theta}_1 = \hat{\beta}_0 + \theta_0$$

as revised estimates of θ . Sometimes $\hat{\beta}_0$ is called the vector of increments. We may now place the revised estimates $\hat{\theta}_1$ in Eq. (12.24) (in the same roles played by the initial estimates θ_0) and then produce another set of revised estimates, say $\hat{\theta}_2$, and so forth.

In general, we have at the k -th iteration

$$\hat{\theta}_{k+1} = \hat{\theta}_k + \hat{\beta}_k = \hat{\theta}_k + (\mathbf{Z}_k' \mathbf{Z}_k)^{-1} \mathbf{Z}_k' (\mathbf{y} - \mathbf{f}_k)$$

where

$$\mathbf{Z}_k = [\mathbf{Z}_{ij}^k]$$

$$\mathbf{f}_k = [f_{1k}, f_{2k}, \dots, f_{nk}]'$$

$$\hat{\theta}_k = [\theta_{1k}, \theta_{2k}, \dots, \theta_{pk}]'$$

This iterative process continues until convergence, that is, until

$$\left[\frac{(\hat{\theta}_{jk+1} - \hat{\theta}_{jk})}{\hat{\theta}_{jk}} \right] < \delta, \quad j = 1, 2, \dots, p$$

where δ is some small number, say 1.0×10^{-6} . At each iteration the residual sum of squares $S(\hat{\theta}_k)$ should be evaluated to ensure that a reduction in its value has been obtained.

Steepest Descent

The method of steepest descent is widely used in response surface methodology to move from an initial estimate of the optimum conditions for a process to a region more likely to contain the optimum. The major disadvantage of this method in solving the nonlinear regression problem is that it may converge very slowly.

Starting Values

Fitting a non-linear regression model requires starting values θ_0 of the model parameters.

Good starting values that is the values of θ_0 that are close to the true parameter values will minimize convergence difficulties.

The following are sources of starting values:

- If you have expert knowledge or prior experience with the data or the underlying process, you can use reasonable guesses for the starting values.
- In some cases, nonlinear models can be linearized through transformations. For example, transforming an exponential model using logarithms can simplify it to a linear form. You can then fit a linear regression model and use the estimates as starting values for the nonlinear regression.
- A grid search or trial-and-error approach involves testing a wide range of plausible starting values, running the algorithm from each set, and selecting the one that leads to the best or fastest convergence.
- Some statistical software packages (such as R, SAS, SPSS, and MATLAB) have default algorithms or heuristics that automatically generate starting values for the parameters. These are typically derived from basic statistical measures (e.g., means, medians, or maximum values) of the input data.

Examples of Non-Linear Regression Models The logistic growth model is

$$y = \frac{\theta_1}{1 + \theta_2 \exp(-\theta_3 x)} + \epsilon$$

The parameters in this model have a simple physical interpretation. For $x = 0$,

$$y = \frac{\theta_1}{(1 + \theta_2)}$$

is the level of y at time (or level) zero. The parameter θ_1 is the limit to growth as $x \rightarrow \infty$. The values of θ_2 and θ_3 must be positive.

The Gompertz model given by

$$y = \theta_1 \exp(-\theta_2 e^{-\theta_3 x}) + \epsilon$$

is another widely used growth model. At $x = 0$, we have

$$y = \theta_1 e^{-\theta_2}$$

and θ_1 is the limit to growth as $x \rightarrow \infty$.

The Weibull growth model is

$$y = \theta_1 - \theta_2 \exp(-\theta_3 x^{\theta_4}) + \epsilon$$

When $x = 0$, we have

$$y = \theta_1 - \theta_2$$

while the limiting growth is θ_1 as $x \rightarrow \infty$.

Example in R Using the data set provided fit the following non-linear regression model:

$$y = \theta_1 e^{\theta_2 x} + \epsilon$$

```
x <- c(0.5,1,2,4,8,9,10)
y <- c(0.68, 0.45,2.50,6.19,56.1,89.8,147.7)
model <- nls(y ~ theta1 * exp(theta2 *x), start = list(theta1 = 10, theta2 = 0.9))
model
```